

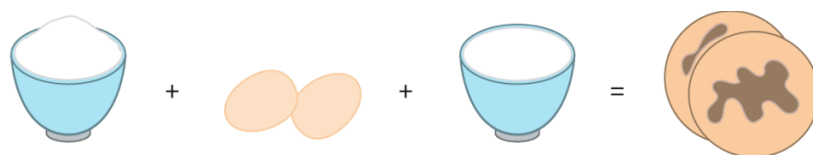
# Reacting ratios

## Learning outcomes

By the end of this section you should be able to:

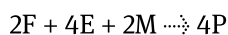
- Calculate reacting masses and/or volumes of reactants and products.
- Calculate the concentrations of reactants and products for reactions occurring in solution.

Recall the recipe for pancakes that we discussed previously where one cup of flour, two small eggs and one cup of milk was mixed and cooked to produce two pancakes. We represented this recipe as the chemical equation as shown in **Figure 1**.

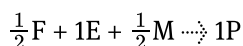


**Figure 1.** Pancake recipe equation.

What would need to be done to the recipe to make four pancakes? You're right, we could double it. This would mean that you would have two cups of flour, four small eggs and two cups of milk. This could then be represented by:

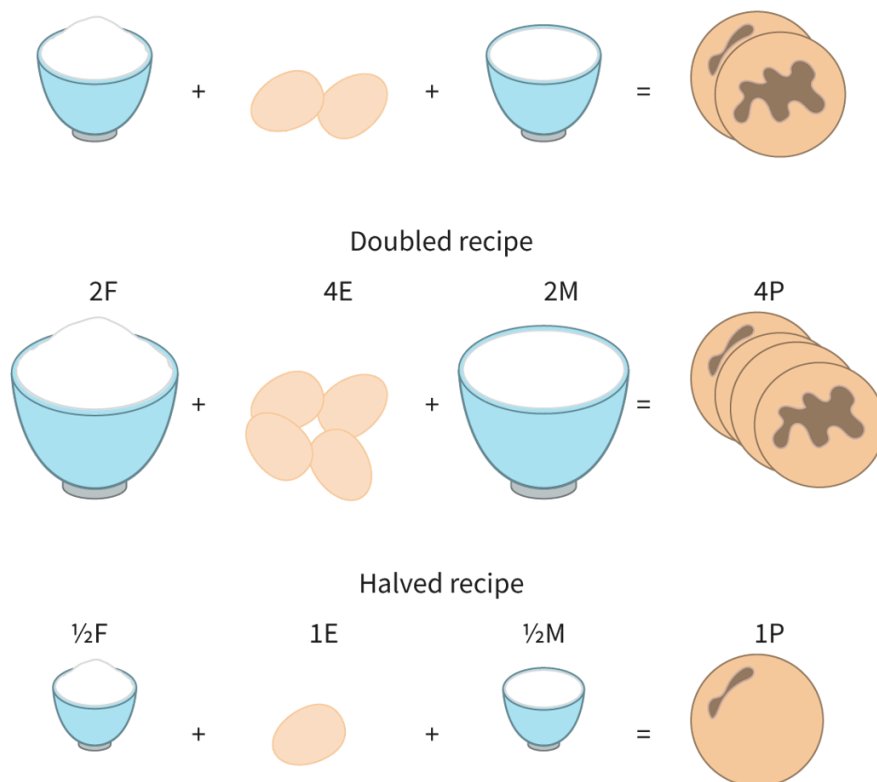


You can see that the ratio of ingredients has not changed. What if you had lots of flour and milk and one egg in total. How many pancakes could you make?



One pancake! If you tried to use a full cup of flour or milk, then the food produced would certainly not taste like a pancake!

We can figure out how much product can be made by looking at the amount of ingredients we have, or we can figure out the amount of ingredients we will need based on the number of pancakes we want to make (**Figure 2**).



**Figure 2.** Recipes can be scaled up or down so long as the ratios stay the same.

More information for figure 2

The image consists of three parts that show a basic recipe, a doubled recipe, and a halved recipe. Each section uses illustrated icons of ingredients, which include bowls labeled with amounts of flour (F) and milk (M), and images of eggs (E), leading to a result in pancakes (P).

**1. Basic Recipe:**

- Shows a simple combination of a bowl of flour, two eggs, and a bowl of milk resulting in two pancakes.

**3. Doubled Recipe:**

- Illustrates twice the amount, with 2F (double the flour), 4E (four eggs), and 2M (double the milk), which produces 4P (four pancakes).

**5. Halved Recipe:**

- Displays half the amount required, with  $\frac{1}{2}$ F (half the flour), 1E (one egg), and  $\frac{1}{2}$ M (half the milk), producing 1P (one pancake).

Each variation is aligned horizontally with simple mathematical symbols to emphasize the scaling, showing that ingredient ratios must remain constant to maintain the recipe's consistency.

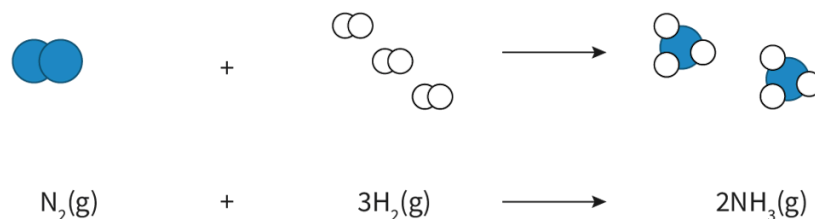
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This same idea can be applied to chemical equations. You can use the ratios in the reaction to determine how much of each reactant is needed or how much product can be created. Similar to how it is important to make sure that a recipe is correct to begin with, it is important to make sure that chemical equations are always balanced before you begin. If this is something you struggle with, you may want to revise [subtopic S1.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) before you begin.

## Reacting ratios

When we write balanced chemical equations, we place coefficients in front of reactants and products. What do these numbers represent? As long as the correct ratios are maintained, we can actually apply any unit of measurement to them.

Take the reaction between nitrogen and hydrogen gases to produce ammonia molecules for example (**Figure 3**).



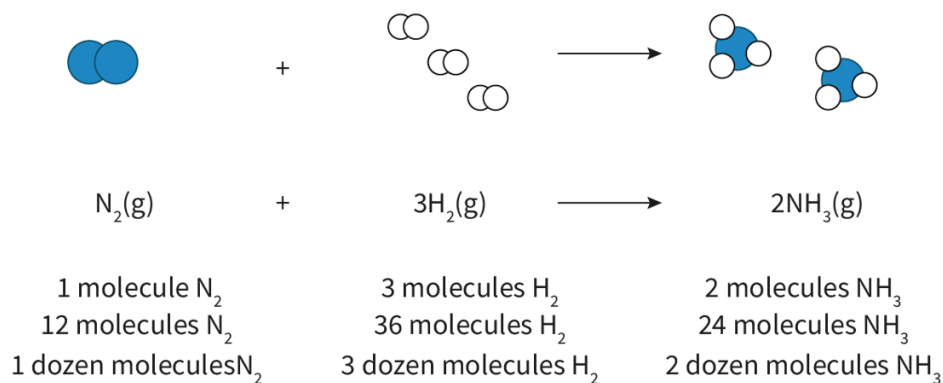
**Figure 3.** The reaction between nitrogen and hydrogen gases to produce ammonia molecules.

 More information for figure 3

This diagram illustrates the chemical reaction between nitrogen ( $\text{N}_2$ ) and hydrogen ( $\text{H}_2$ ) gases to form ammonia ( $\text{NH}_3$ ). On the left side, there is a depiction of one nitrogen molecule ( $\text{N}_2$ ), represented as two blue circles connected, indicating a diatomic nitrogen gas. Next to it, three hydrogen molecules ( $\text{H}_2$ ) are shown as pairs of smaller white circles. An arrow points from these reactants to the right side, illustrating the formation of ammonia molecules ( $\text{NH}_3$ ). Two ammonia molecules are depicted as structures with a central blue circle (indicating nitrogen) surrounded by three smaller white circles (indicating hydrogen). Below this visual representation, the equation is expressed as:  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ .

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The ratios could represent the number of molecules (**Figure 4**):



**Figure 4.** The reaction between nitrogen and hydrogen gases to produce ammonia represented using molecules.

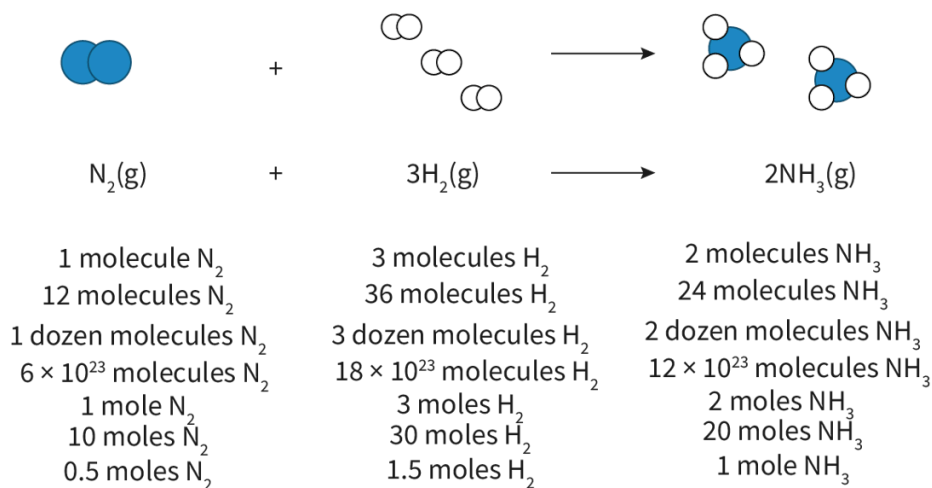
More information for figure 4

The image illustrates the chemical reaction between nitrogen ( $\text{N}_2$ ) and hydrogen ( $\text{H}_2$ ) gases to produce ammonia ( $\text{NH}_3$ ). It depicts molecules of nitrogen and hydrogen combining to form ammonia molecules. On the left, the image shows a pair of blue circles labeled as one molecule of nitrogen ( $\text{N}_2$ ). Adjacent to it are three pairs of white circles representing three molecules of hydrogen ( $\text{H}_2$ ). An arrow points towards the right signifying the reaction, leading to the creation of two ammonia molecules, represented by blue and white linked circles. Below the diagram are proportional representations: 1 molecule  $\text{N}_2$ , 3 molecules  $\text{H}_2$ , resulting in 2 molecules  $\text{NH}_3$ ; 12 molecules  $\text{N}_2$ , 36 molecules  $\text{H}_2$ , resulting in 24 molecules  $\text{NH}_3$ ; 1 dozen molecules  $\text{N}_2$ , 3 dozen molecules  $\text{H}_2$ , producing 2 dozen molecules  $\text{NH}_3$ .

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However, trying to measure out the number of molecules would be quite difficult. One hydrogen molecule has a mass of  $3.4 \times 10^{-24}$  g or 0.0000000000000000000000034 g, which is not very easy to measure.

You will recall learning about the mole as a unit of measurement in [subtopic S1.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/). Moles are a convenient way to measure the number of atoms and so the coefficients used to balance chemical equations actually represent moles (**Figure 5**).



**Figure 5.** The use of moles to quantify reactants and products.

More information for figure 5

This image is a diagram illustrating a chemical equation involving nitrogen ( $\text{N}_2$ ), hydrogen ( $\text{H}_2$ ), and ammonia ( $\text{NH}_3$ ). It visually represents the reaction of nitrogen gas with hydrogen gas to produce ammonia gas. The left side of the equation shows 1 molecule of  $\text{N}_2$  combining with 3 molecules of  $\text{H}_2$ . Different representations of these molecules are included: molecules, dozens of molecules, Avogadro's number ( $6 \times 10^{23}$  and its multiples), and moles. The right side of the equation displays the products, 2 molecules of  $\text{NH}_3$ , along with their equivalent representations including dozens, Avogadro's number, and moles. Arrows indicate the direction of the chemical reaction from reactants to products.

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You may find it helpful to practise converting mole ratios in a chemical equation before you continue, and don't forget what you learned about Avogadro's law of combining volumes in [section S1.4.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/).

Worked example 1 shows this in practice.

## Worked example 1

Each reaction has a specific mole ratio.

Deduce the missing of the coefficients when the amount of reactant changes.

### Question 1

|                   |          |   |              |               |     |
|-------------------|----------|---|--------------|---------------|-----|
| Chemical Equation | 4Na      | + | $\text{O}_2$ | $\rightarrow$ | 2Na |
| Mole Ratio        | 24.0 mol |   |              |               |     |

|       |           |  |  |  |  |
|-------|-----------|--|--|--|--|
| (mol) | 0.100 mol |  |  |  |  |
|-------|-----------|--|--|--|--|

Question 2

|                   |                  |   |                 |   |                 |   |                   |
|-------------------|------------------|---|-----------------|---|-----------------|---|-------------------|
| Chemical Equation | 4NH <sub>3</sub> | + | 3O <sub>2</sub> | → | 2N <sub>2</sub> | + | 6H <sub>2</sub> O |
| Mole Ratio (mol)  | 0.500 mol        |   |                 |   |                 |   |                   |
|                   | 10.0 mol         |   |                 |   |                 |   |                   |

Question 1

|                                                                                                                                           |                                                                                                                                                                |   |                |  |                   |     |   |                |                  |   |   |   |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|--|-------------------|-----|---|----------------|------------------|---|---|---|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s                                                                             | Calculations                                                                                                                                                   |   |                |  |                   |     |   |                |                  |   |   |   |
| S<br>t<br>e<br>p<br>1:<br>I<br>d<br>e<br>n<br>t<br>i<br>f<br>y<br>t<br>h<br>e<br>s<br>i<br>m<br>p<br>l<br>e<br>s<br>t<br>w<br>h<br>o<br>l | <table><tr><td>Chemical Equation</td><td>4Na</td><td>+</td><td>O<sub>2</sub></td></tr><tr><td>Mole Ratio (mol)</td><td>4</td><td>:</td><td>1</td></tr></table> |   |                |  | Chemical Equation | 4Na | + | O <sub>2</sub> | Mole Ratio (mol) | 4 | : | 1 |
| Chemical Equation                                                                                                                         | 4Na                                                                                                                                                            | + | O <sub>2</sub> |  |                   |     |   |                |                  |   |   |   |
| Mole Ratio (mol)                                                                                                                          | 4                                                                                                                                                              | : | 1              |  |                   |     |   |                |                  |   |   |   |

|                                                                                                                                                         |                   |           |   |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------|---|----------------|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s                                                                                           | Calculations      |           |   |                |
| e<br>n<br>u<br>m<br>b<br>e<br>r<br>r<br>a<br>t<br>i<br>o<br>.                                                                                           |                   |           |   |                |
| S<br>t<br>e<br>p<br>2<br>:<br>A<br>p<br>p<br>l<br>y<br>t<br>h<br>e<br>r<br>a<br>t<br>i<br>o<br>w<br>i<br>t<br>h<br>n<br>e<br>w<br>c<br>o<br>e<br>f<br>f |                   |           |   |                |
|                                                                                                                                                         | Chemical Equation | 4Na       | + | O <sub>2</sub> |
|                                                                                                                                                         | Mole Ratio (mol)  | 4         | : | 1              |
|                                                                                                                                                         |                   | 24.0 mol  | : | 6.00 m         |
|                                                                                                                                                         |                   | 0.100 mol | : | 0.025 n        |

|                                                               |              |
|---------------------------------------------------------------|--------------|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s | Calculations |
| ci<br>e<br>n<br>t.                                            |              |

**Question 2**

|                                                                                                                             |                                                                                                                                                                                               |   |                 |   |  |                   |                  |   |                 |   |                  |   |  |   |  |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|---|--|-------------------|------------------|---|-----------------|---|------------------|---|--|---|--|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s                                                               | Calculations                                                                                                                                                                                  |   |                 |   |  |                   |                  |   |                 |   |                  |   |  |   |  |
| S<br>t<br>e<br>p<br>1:<br>Id<br>e<br>n<br>t<br>i<br>f<br>y<br>t<br>h<br>e<br>s<br>i<br>m<br>p<br>l<br>e<br>s<br>t<br>w<br>h | <table><tr><td>Chemical Equation</td><td>4NH<sub>3</sub></td><td>+</td><td>3O<sub>2</sub></td><td>→</td></tr><tr><td>Mole Ratio (mol)</td><td>4</td><td></td><td>3</td><td></td></tr></table> |   |                 |   |  | Chemical Equation | 4NH <sub>3</sub> | + | 3O <sub>2</sub> | → | Mole Ratio (mol) | 4 |  | 3 |  |
| Chemical Equation                                                                                                           | 4NH <sub>3</sub>                                                                                                                                                                              | + | 3O <sub>2</sub> | → |  |                   |                  |   |                 |   |                  |   |  |   |  |
| Mole Ratio (mol)                                                                                                            | 4                                                                                                                                                                                             |   | 3               |   |  |                   |                  |   |                 |   |                  |   |  |   |  |



|                                                                                                                                                    |                   |                              |   |                 |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|---|-----------------|---|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s                                                                                      | Calculations      |                              |   |                 |   |
| o<br>l<br>e<br>n<br>u<br>m<br>b<br>e<br>r<br>r<br>a<br>t<br>i<br>o<br>.                                                                            |                   |                              |   |                 |   |
| S<br>t<br>e<br>p<br>2<br>:<br>A<br>p<br>p<br>l<br>y<br>t<br>h<br>e<br>r<br>a<br>t<br>i<br>o<br>w<br>i<br>t<br>h<br>n<br>e<br>w<br>c<br>o<br>e<br>f | Chemical Equation | 4NH <sub>3</sub>             | + | 3O <sub>2</sub> | → |
|                                                                                                                                                    | Mole Ratio (mol)  | 4                            |   | 3               |   |
|                                                                                                                                                    |                   | 0.500<br>mol NH <sub>3</sub> | : | 0.375<br>mol    | : |
|                                                                                                                                                    |                   | 10.0<br>mol NH <sub>3</sub>  | : | 7.50<br>mol     | : |

|                                                               |              |
|---------------------------------------------------------------|--------------|
| S<br>o<br>l<br>u<br>t<br>i<br>o<br>n<br>s<br>t<br>e<br>p<br>s | Calculations |
| fi<br>ci<br>e<br>n<br>t.                                      |              |

Now try to apply this concept in this next question.



### Exercise 1

Click a question to answer



1. Fill in the blanks with an appropriate value to reflect the simplest whole number ratio.

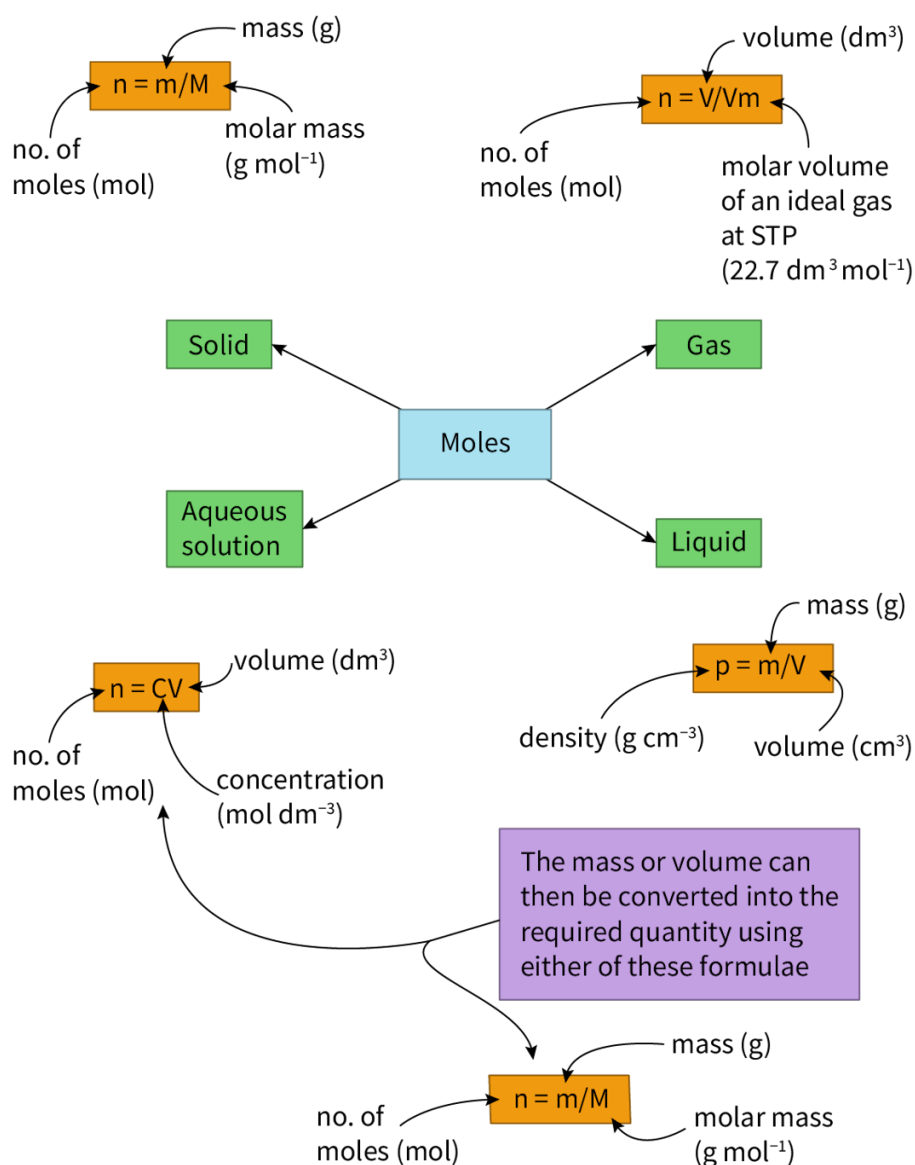
|                   |          |   |                  |   |                    |
|-------------------|----------|---|------------------|---|--------------------|
| Chemical Equation | 2Al      | + | 3Cl <sub>2</sub> | → | 2AlCl <sub>3</sub> |
| Mole Ratio (mol)  | 2        | : | _____            | : | 2                  |
|                   | 5.0 mol  | : | _____            | : | 5.0 mol            |
|                   | 0.10 mol | : | _____            | : | 0.10 mol           |

✓ Check answers

# Moles and determining mass, concentration or volume

Mole ratios are especially useful as they allow you to easily convert the mole values to a mass, volume or concentration of a particular chemical species, depending on whether it is a solid, liquid, gas or an aqueous solution.

You may recall from [subtopic S1.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) that the relationship between moles, mass, concentration and volume are represented by formulae can be found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) of the DP chemistry data booklet. The molar volume of an ideal gas,  $22.7 \text{ dm}^3 \text{ mol}^{-1}$ , can be found in section 2 (**Figure 6**).



**Figure 6.** The relationship between moles, mass, concentration and volume.

More information for figure 6

The diagram is a flowchart illustrating the relationships between moles, mass, concentration, and volume. At the center of the diagram, there is a blue box labeled 'Moles.' From this central box, arrows point to and from various boxes:

- To the top-left, a green box labeled 'Solid' is connected to the 'Moles' box.
- Below 'Solid,' an orange box states the formula ' $n = m/M$ ,' where ' $n$ ' represents the number of moles in moles (mol), ' $m$ ' the mass in grams (g), and ' $M$ ' the molar mass in grams per mole ( $\text{g mol}^{-1}$ ).
- To the top-right, an orange box shows ' $n = V/V_m$ ,' where ' $V$ ' is volume in cubic decimeters ( $\text{dm}^3$ ), and ' $V_m$ ' is the molar volume of an ideal gas at STP ( $22.7 \text{ dm}^3 \text{ mol}^{-1}$ ).
- Next to it, a green box labeled 'Gas' is also connected to the 'Moles' box.
- A green box labeled 'Liquid' is directly connected 'Moles.'
- Another orange box, ' $\rho = m/V$ ,' involves ' $\rho$ ' as density in grams per cubic centimeter ( $\text{g cm}^{-3}$ ) coupled with ' $m$ ' mass and ' $V$ ' volume in cubic centimeters ( $\text{cm}^3$ ).
- Finally, an orange box illustrates ' $n = CV$ ,' leveraging ' $C$ ' concentration in moles per cubic decimeter ( $\text{mol dm}^{-3}$ ) and ' $V$ ' for volume.
- At the bottom right, a purple box notes, 'The mass or volume can then be converted into the required quantity using either of these formulae.'

Overall, the diagram visually explains how different states of matter (solid, liquid, gas) and various formulas connect to the concept of moles.

[Generated by AI]

## Practical skills

**Tool 1:** Experimental techniques — Measuring variables

**Inquiry 2:** Collecting and processing data — Collecting data

Although they are typically measured by volume, liquids can also be measured using their mass. The moles of liquid are then calculated by using the density of that liquid, and whatever other information is available.

For example, water has a density of  $1.000 \text{ g cm}^{-3}$ . If a student measured  $18.00 \text{ g}$  of water then they can identify that it would have a volume of  $18.00 \text{ cm}^3$ .

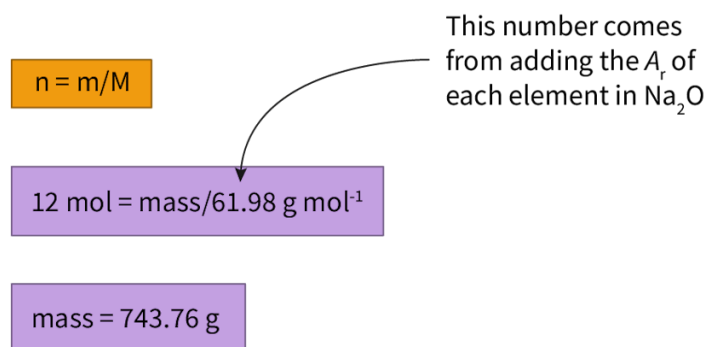
However, not every liquid has a density of  $1.000 \text{ g cm}^{-3}$ , so volume is not equal to mass for all liquids.

For example, the reaction for sodium and oxygen is given in the worked example above. It is possible to determine the mass of sodium oxide produced from 24.0 moles of sodium.

24.0 moles of sodium react with 6.00 moles of oxygen to produce 12.0 moles of sodium oxide. What is the mass of 12.0 moles of sodium oxide,  $\text{Na}_2\text{O}$ ?

|                   |              |   |              |               |                        |
|-------------------|--------------|---|--------------|---------------|------------------------|
| Chemical Equation | $4\text{Na}$ | + | $\text{O}_2$ | $\rightarrow$ | $2\text{Na}_2\text{O}$ |
| Mole Ratio (mol)  | 24.0 mol     | : | 6.00 mol     | :             | 12.0 mol               |

Determine the  $M$  of  $\text{Na}_2\text{O}$  using the periodic table and input the values into the formula (**Figure 7**).



**Figure 7.** Finding the mass of sodium oxide.

 More information for figure 7

The diagram illustrates the calculation of the mass of sodium oxide using a formula. It starts with a formula box labeled ' $n = m/M$ ' in orange, where ' $n$ ' is moles, ' $m$ ' is mass, and ' $M$ ' is molar mass. Below this, a purple box shows the equation ' $12 \text{ mol} = \text{mass} / 61.98 \text{ g mol}^{-1}$ ', where  $61.98 \text{ g mol}^{-1}$  is the molar mass derived from adding the relative atomic mass ( $A_r$ ) of each element in  $\text{Na}_2\text{O}$ . The molar mass calculation is explained in a text box pointing to the equation, stating, 'This number comes from adding the  $A_r$  of each element in  $\text{Na}_2\text{O}$ '. Below this, another purple box shows ' $\text{mass} = 743.76 \text{ g}$ ', indicating that the mass is 743.76 grams after multiplying the molar mass by the moles.

[Generated by AI]

What would happen if you were not given the moles of sodium above but were instead given a mass of sodium?

Of course, you could just convert the mass into moles using the same formula.

### Study skills

The values for  $A_r$  given in [section 7 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet are to two decimal places and so these should be used in calculations.

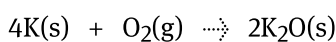
In the lab we always measure in mass, or volume and concentration so you can expect to have to first convert this into moles before you can apply the mole ratio. The following steps summarise how to determine the mass, volume or concentration of an unknown reactant or product:

1. Balance the equation. Identify the known reactant/product and unknown reactant/product.
2. Convert the 'known' mass, concentration or volume given the question into moles.
3. Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.
4. Convert the moles value into the required quantity (mass, concentration or volume). Remember – if the question asks for 'amount' of something then it is acceptable to leave the answer in moles.

The next worked example will demonstrate how this can be done for a reaction involving solids.

## Worked example 2

Calculate the mass of potassium oxide formed when 7.82 g of potassium is burned in oxygen, according to the following chemical equation.

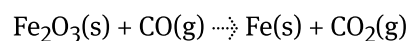


| Solution steps                                                                          | Calculations                                   |
|-----------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Step 1:</b> Identify the 'known' reactant/product and unknown reactant/product.      | $4\text{K(s)} + \text{O}_2\text{(g)}$<br>known |
| <b>Step 2:</b> Convert the mass, concentration or volume given the question into moles. | For $\text{K(s)}$                              |

| Solution steps                                                                                                 | Calculations                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                | Mass, $m$ : 7.82 g<br>$M = 39.10 \text{ g}$<br>$n = \frac{m}{M}$ $= \frac{7.82}{39.10}$ $= 0.200 \text{ mol}$                                                 |
| <b>Step 3:</b> Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value. | $4\text{K(s)} : 2\text{K}_2\text{CO}_3(\text{s})$<br>$4 : 2$<br>$0.200 \text{ mol K}$                                                                         |
| <b>Step 4:</b> Convert the moles value into the required quantity (mass, concentration or volume).             | For $\text{K}_2\text{O(s)}$<br>$n = 0.100 \text{ mol}$<br>$M = 94.20 \text{ g}$<br>$m = ?$<br>$n = \frac{m}{M}$ $0.100 = \frac{m}{94.2}$ $m = 9.42 \text{ g}$ |

### Worked example 3

Calculate the mass (in kg) of iron produced when 32.0 kg of iron(III) oxide is heated with excess CO, according to this unbalanced chemical equation:



| Solution steps                                                                                                                                                  | Calculations                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1:</b> Balance the equation. Identify the known reactant/product and unknown reactant/product.                                                          | $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g})$<br>known                                                                                                                                                             |
| <b>Step 2:</b> Convert the mass, concentration or volume given the question into moles. Don't forget to convert units if required, for example, from kg into g. | For $\text{Fe}_2\text{O}_3(\text{s})$<br><br>Mass, $m$ : 32.0 kg<br>(note the unit conversion)<br><br>$M = 159.70 \text{ g mol}^{-1}$<br><br>$n = \frac{m}{M}$ $= \frac{32000}{159.70}$ $= 200.4 \text{ mol}$                 |
| <b>Step 3:</b> Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.                                                  | $\text{Fe}_2\text{O}_3(\text{s}) : 2\text{Fe}(\text{s})$<br>1 : 2<br><br>200.4 mol $\text{Fe}_2\text{O}_3$                                                                                                                    |
| <b>Step 4:</b> Convert the moles value into the required quantity (mass, concentration or volume).                                                              | For $\text{Fe}(\text{s})$<br><br>$n = 400.8 \text{ mol}$<br><br>$M = 55.85 \text{ g mol}^{-1}$<br><br>$m = ?$<br><br>$n = \frac{m}{M}$ $400.8 = \frac{m}{55.85}$<br><br>$m = 22\,385 \text{ g of Fe}$<br><br>So 22.4 kg of Fe |

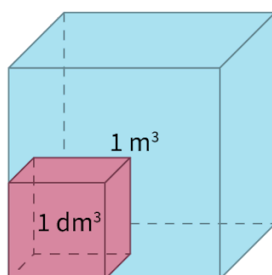
For a reaction involving aqueous solutions, this process is essentially the same, but now the moles will be converted using  $n = CV$  where  $C$  is the concentration and  $V$  is the volume of the aqueous solution. This is because the quantity of aqueous solution is measured in volume, rather than mass.



## Study skills

You learned in [section S1.4.5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) that it is important to be careful with units when dealing with volumes of liquids or gases. You must be able to convert easily between  $\text{m}^3$ ,  $\text{dm}^3$  and  $\text{cm}^3$ . This conversion is also given for you in [section 4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/>) of the DP chemistry data booklet.

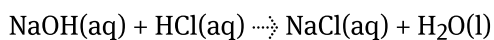
$$\begin{array}{ccccc}
 1 \text{ dm}^3 = 1 \times 10^{-3} \text{ m}^3 & & 1000 \text{ dm}^3 = 1 \text{ m}^3 & & \\
 1 \text{ cm}^3 = 1 \times 10^{-3} \text{ dm}^3 & & 1000 \text{ cm}^3 = 1 \text{ dm}^3 & & \\
 \text{Divide by 1000} & & \text{Divide by 1000} & & \\
 \text{cm}^3 & \xleftrightarrow{\text{Multiply by 1000}} & \text{dm}^3 & \xleftrightarrow{\text{Multiply by 1000}} & \text{m}^3
 \end{array}$$



📖 More information

## Worked example 4

A reaction occurs when sodium hydroxide and hydrochloric acid are mixed, as shown in this balanced chemical equation:



$23.75 \text{ cm}^3$  of NaOH, with a concentration of  $0.01000 \text{ mol dm}^{-3}$ , was required to react completely with  $25.00 \text{ cm}^3$  of HCl. Determine the concentration of HCl.

| Calculations                                                                                          | Calculations                                                                                    |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Step 1:</b> Balance the equation. Identify the known reactant/product and unknown reactant/product | $\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$<br>known |

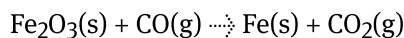
| Calculations                                                                                                   | Calculations                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 2:</b> Convert the mass, concentration or volume given the question into moles.                        | <p>For NaOH(aq)</p> $V = 23.75 \text{ cm}^3$ $= 0.02375 \text{ dm}^3$ <p>(remember to convert)</p> $C = 0.01000 \text{ mol dm}^{-3}$ $n = ?$ $n = CV$ $= 0.01000 \times 0.02375$ $= 0.0002375 \text{ mol}$                 |
| <b>Step 3:</b> Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value. | <p>NaOH(aq) : HCl(aq)</p> <p>1 : 1</p> $0.0002375 : x \text{ mol HCl}$ $x = 0.0002375 \text{ mol}$                                                                                                                         |
| <b>Step 4:</b> Convert the moles value into the required quantity (mass, concentration or volume).             | <p>For HCl(aq)</p> $n = 0.0002375 \text{ mol}$ $V = 25.00 \text{ cm}^3$ $= 0.025 \text{ dm}^3$ $C = ?$ $n = CV$ $0.0002375 = C \times 0.025$ $C = 0.009500 \text{ mol dm}^{-3}$ $9.500 \times 10^{-3} \text{ mol dm}^{-3}$ |

The questions above were focused separately on solids and aqueous solutions. However, it is quite normal to have to use a combination of the mole conversion formulas for solids, gases and aqueous solutions to determine the final answer.

Let's revisit the equation for worked example 3 where we calculated the mass of pure iron produced. Carbon dioxide is also produced in that reaction. How could we calculate the volume of carbon dioxide produced? The next worked example explores that, but it would be good practice to try to come up with your own answer before you go through the worked solutions.

## Worked example 5

The reaction between 32.00 kg of iron(III) oxide and excess carbon monoxide to produce iron and carbon dioxide was carried out at standard temperature and pressure (STP) with the following unbalanced reaction equation.



Determine the volume of carbon monoxide required for the reaction.

| Solution steps                                                                                                 | Calculations                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1:</b> Balance the equation. Identify the known reactant/product and unknown reactant/product.         | $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 2\text{Fe}$<br>unknown                                                                                                                                             |
| <b>Step 2:</b> Convert the mass, concentration or volume given the question into moles.                        | For $\text{Fe}_2\text{O}_3(\text{s})$<br><br>mass: 32.00 kg = 32 000<br>(note the unit conversion)<br><br>$M = 159.70 \text{ g mol}^{-1}$<br><br>$n = ?$<br><br>$n = \frac{m}{M}$<br>$= \frac{32000}{159.70}$<br>$= 200.4 \text{ mol}$ |
| <b>Step 3:</b> Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value. | $\text{Fe}_2\text{O}_3(\text{s}) : 3\text{CO}(\text{g})$<br>1 : 3<br><br>200.4 mol $\text{Fe}_2\text{O}_3 : x \text{ mol}$<br>$x = 6$                                                                                                  |
| <b>Step 4:</b> Convert the moles value into the required quantity (mass, concentration or volume).             | For $\text{CO}(\text{g})$<br><br>$n = 601.2 \text{ mol}$<br><br>$V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$<br><br>$V = ?$                                                                                                              |

| Solution steps | Calculations                                                                                                              |
|----------------|---------------------------------------------------------------------------------------------------------------------------|
|                | $n = \frac{V}{V_m}$ $601.2 = \frac{V}{22.7}$ $V = 13\,647 \text{ dm}^3$ $V = 1.365 \times 10^4 \text{ dm}^3 \text{ of C}$ |

The reaction above required three moles of carbon monoxide which is  $68.1 \text{ dm}^3$  of gas. What volume of carbon dioxide would also have been produced from the reaction? Recall what you learned about Avogadro's law in [section S1.4.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/>) to help you answer this question.

As both gases have the same coefficient and thus the same number of moles, that means they would have the same volume. Remember that the volume of the gas is independent of what the gas is – it is a molar quantity. However, it is important that the reaction conditions must stay constant as a change in temperature or pressure would impact the volume of the gas.

## Worked example 6

What volume of carbon dioxide,  $\text{CO}_2$ , in  $\text{cm}^3$ , is produced when  $50.0 \text{ cm}^3$  of  $1.00 \text{ mol dm}^{-3}$  hydrochloric acid,  $\text{HCl}$ , reacts completely with excess calcium carbonate,  $\text{CaCO}_3$ ?



| Solution steps                                                                                         | Calculations                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1:</b> Balance the equation. Identify the known reactant/product and unknown reactant/product. | $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{a})$ <p style="text-align: center;">known</p> <p style="text-align: center;">unknown</p> |
| <b>Step 2:</b> Convert the mass, concentration or volume given the question into moles.                | <p>For <math>\text{HCl}(\text{aq})</math></p> <p>Volume: <math>50.0 \text{ cm}^3 = 0.0500 \text{ dm}^3</math></p> <p><math>C = 1.00 \text{ mol dm}^{-3}</math></p>         |

| Solution steps                                                                                                 | Calculations                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                | $n = ?$<br><br>$n = CV$<br><br>$n = 1.00 \text{ mol dm}^{-3} \times 0.0500 \text{ dm}^3$                                                                                                                                                                                                |
| <b>Step 3:</b> Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value. | $\text{HCl(aq)} : \text{CO}_2(\text{g})$<br>$2 : 1$<br><br>$0.0500 \text{ mol HCl} : x \text{ mol CO}_2$<br>$x = 0.0250 \text{ mol}$                                                                                                                                                    |
| <b>Step 4:</b> Convert the moles value into the required quantity (mass, concentration or volume).             | For $\text{CO}_2(\text{g})$<br><br>$n = 0.0250 \text{ mol}$<br><br>$V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$<br><br>$V = ?$<br><br>$n = \frac{V}{V_m}$<br><br>$0.0250 = \frac{V}{22.7}$<br><br>$V = 0.568 \text{ dm}^3$<br><br>$V = 568 \text{ cm}^3 \text{ of CO}_2 \text{ produced}$ |

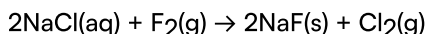
The next activity will allow you an opportunity to practise some of these calculations using experimental data. Remember to consider the physical state of the substances involved before you begin and pay particular attention to units while you complete them.

### Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

### Questions

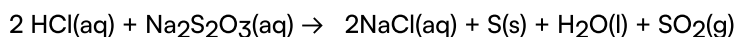
1. Sodium fluoride can be used to treat tooth decay. It is produced in a reaction between fluorine, a pale yellow-green gas, and sodium chloride solution. Chlorine gas is also produced in this reaction, which can be represented by the following balanced equation:



A student wanted to prepare 84.00 g of sodium fluoride according to this reaction.

Calculate the volume of fluorine gas which would be needed to do this.

2. A student wanted to determine the concentration of hydrochloric acid by reacting it with sodium thiosulphate solution. They formulated a balanced chemical equation for the reaction:



16.7 cm<sup>3</sup> of a 0.100 mol dm<sup>-3</sup> sodium thiosulphate solution was required to react with 20.0 cm<sup>3</sup> of the hydrochloric acid.

Determine the concentration of the hydrochloric acid solution in mol dm<sup>-3</sup>.

3. Calcium carbonate can be decomposed to produce calcium oxide which is used in the production of glass, cement and agricultural products. The reaction also produces carbon dioxide gas, according to the balanced equation:

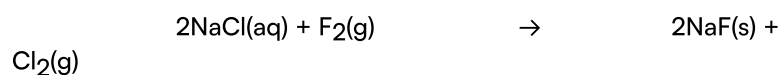


While decomposing calcium carbonate at STP, a student produced 100.00 g of calcium oxide.

Determine the volume of carbon dioxide gas also produced by the reaction in dm<sup>3</sup>.

#### Answer 1.

**Step 1:** Check the equation is balanced

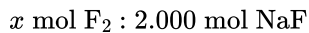
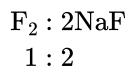


| Substance | For F <sub>2</sub>                                               |
|-----------|------------------------------------------------------------------|
| Property  | $V = ?$<br>$V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$<br>$n = ?$ |

**Step 2:** Convert the mass into moles of NaF

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{84.00}{41.99} \\
 &= 2.000 \text{ mol}
 \end{aligned}$$

**Step 3:** Use the mole ratio to determine the moles of the unknown value



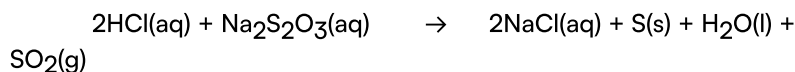
$$x = 1.000 \text{ mol F}_2$$

**Step 4:** Convert the moles value into volume of  $\text{F}_2$

$$\begin{aligned}
 n &= \frac{V}{V_m} \\
 1.000 \text{ mol} &= \frac{V}{22.7} \\
 V &= 22.7 \text{ dm}^3
 \end{aligned}$$

**Answer 2.**

**Step 1:** Check the equation is balanced

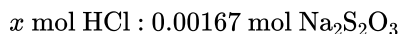
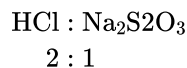


| Substance | For HCl                                       |            |
|-----------|-----------------------------------------------|------------|
| Property  | $n = ?$<br>$c = ?$<br>$V = 20.0 \text{ cm}^3$ | $r$<br>$($ |

**Step 2:** Convert the concentration and volume into moles of  $\text{Na}_2\text{S}_2\text{O}_3$

$$\begin{aligned}
 n &= C \times V \\
 &= 0.100 \times 0.0167 \\
 &= 0.00167 \text{ mol}
 \end{aligned}$$

**Step 3:** Use the mole ratio to determine the moles of the unknown value



$$x = 0.00334 \text{ mol of HCl}$$

**Step 4:** Convert the moles value into concentration of HCl

$$\begin{aligned}
 n &= C \times V \\
 3.34 \times 10^{-3} \text{ mol} &= C \times 0.0200 \text{ dm}^3 \\
 C &= 0.167 \text{ mol dm}^{-3}
 \end{aligned}$$

**Answer 3.**

**Step 1:** Check the equation is balanced



| Substance | For CaO                                                             |
|-----------|---------------------------------------------------------------------|
| Property  | $n = ?$<br>$m = 100.00 \text{ g}$<br>$M = 56.08 \text{ g mol}^{-1}$ |

**Step 2:** Convert the mass of CaO into moles.

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{100.00}{56.08} \\
 &= 1.783 \text{ mol}
 \end{aligned}$$

**Step 3:** Use the mole ratio to determine the moles of CO<sub>2</sub> produced

Mole ratio of CaO to CO<sub>2</sub> is 1:1

Amount of CO<sub>2</sub> produced = 1.783 mol

**Step 4:**

Convert the moles value into volume of CO<sub>2</sub>

$$\begin{aligned}
 n &= \frac{V}{V_m} \\
 1.783 \text{ mol} &= \frac{V}{22.7} \\
 V &= 40.5 \text{ dm}^3
 \end{aligned}$$